Assignment 5
Due at 11:59 pm on Wednesday, October 14

Reading: Sections 6.1 - 6.9 and 8.1 - 8.2 of the textbook.

1. (20 points) For each of the following two player games find all Nash equilibria. In each payoff matrix below the rows correspond to player A's strategies and the columns correspond to player B's strategies. The first entry in each box is player A's payoff and the second entry is player B's payoff.

a)

		Player B L	Player B R
Player A	U	$8,4$	$5,5$
	D	$3,3$	$4,8$

b)

		Player B L	Player B R
Player A	U	$0,0$	$-1,1$
	D	$-1,1$	$2,-2$

c)

		Player B L	Player B R
Player A	U	$1,1$	$3,2$
	D	$0,3$	$4,4$

d)

		Player B L	Player B R
Player A	U	$5,6$	$0,10$
	D	$4,4$	$2,2$

2. (21 Points) Consider the following payoff matrix, where *x* could represent any number.

		Player 2 **L**	Player 2 **R**
Player 1	**U**	*x*, 3	4, 6
	D	2, 9	3, 8

   a. For what values of *x,* if any, does the payoff matrix have a pure-strategy Nash equilibrium at (D,L)?
   b. For what values of *x*, if any, does the payoff matrix have a pure-strategy Nash equilibrium at (U,L)?
   c. For what values of *x*, if any, does the payoff matrix have a mixed-strategy Nash equilibrium at (¼,⅔)?
   d. For what values of *x*, if any, does the payoff matrix have a mixed-strategy Nash equilibrium at (⅓,⅔)?
   e. For what values of *x*, if any, does the payoff matrix have no mixed-strategy Nash equilibrium?
   f. For what values of *x*, if any, is player 1's expected payoff 3 when the players play the mixed strategies (0.7,0.8)?

Show all your work for all parts. If there are values of *x* that satisfy the properties, be sure to (i) list all values of *x* that satisfy the listed properties, (ii) explain why the values of *x* you specified

satisfy the listed properties, and (iii) explain why no other values of *x* would satisfy the listed properties. If no values of *x* satisfy the properties, explain why.
Note: Please be sure to specify all values of x that satisfy the properties specified in the subparts. In some cases, it may not work for any value of x and you need to explain why. In some cases, this may be a range or multiple ranges of values instead of just one number – and you'll need to provide the entire range to get full credit.

3. (22 points) Two drivers are heading straight toward each other. Simultaneously and independently, each one chooses either to Swerve or to go Straight. If both swerve, nothing happens and both get a payoff of 0. If exactly one swerves, the driver who swerved is called a chicken and gets a payoff of -1, while the driver who went straight is called brave and gets a payoff of 1. If neither swerves, the cars crash and both drivers get a payoff of -k, where k is a number greater than 1.
    a. Draw the payoff matrix for this game. Payoffs should be in terms of k.
    b. Find all pure-strategy Nash equilibria. Explain why the pure-strategy Nash equilibria you found do not vary as long as $k > 1$.
    c. Find the mixed-strategy Nash equilibrium. Let p be the probability that Driver 1 swerves and q the probability that Driver 2 swerves. Express your answer in terms of k.
    d. What is the probability that the two cars crash at the mixed-strategy Nash equilibrium? Express your answer in terms of k.
    e. As *k* increases, what happens to the probability that both players will choose to go straight and crash? In the context of what the value *k* represents, explain why increasing *k* changes the probability of crashing in this way.

4. (19 Points) A zero-sum game describes a relationship, competition, or business deal where one person's gain is the other person's loss. In this problem, we will explore how to model zero-sum games using the frameworks from Chapter 6 in the textbook. Consider the zero-sum game where players are trying to guess the lowest number: First, Player 1 privately chooses an even integer between 0 and 999, and Player 2 privately chooses an odd integer between 0 and 999; in this first step, the players don't know each others' choice of integer. Next, both players announce their number. Finally, the player who chose the lower number wins and the player who chose the higher number loses; the loser pays the winner their announced number (i.e., if Player 1 chooses 4 and Player 2 chooses 11, then Player 2 loses and pays Player 1 $4) in dollars.
    a. First, let's model this scenario as a game with two players. Each player's strategies are all the numbers they are allowed to guess, so Player 1's strategies are 0, 2, …, and 998 and Player 2's strategies are 1, 3, …, and 999. Each player's payoff is the number of dollars they gained (as a positive number) or lost (as a negative number since both players want to gain money) in the final step. Fill in the following payoff matrix with each player's payoffs for each choice of strategies in the game:

Player 2

Player 1	1	3	5	…	997	999
0				…		
2				…		
4				…		
…	…	…	…	…	…	…
996				…		
998				…		

b. Find all pure-strategy Nash equilibria of the payoff matrix in part (a). Show all of your work.

c. Next, we'll explore mixed strategies in a similar game. To make calculations easier, let's reduce the number of strategies in the previous game. We'll assume Player 1 is choosing between 0 and some even integer *k*, and Player 2 is choosing between 1 and the odd integer *k*+1. And the payoffs follow the same rules as the game in part b. Fill in the payoffs in the payoff matrix below, and explain why this game has no mixed-strategy Nash equilibria.

Player 2

Player 1	1	k+1
0		
k		

5. (18 Points) Alice and Bob plan to meet after class, but they forgot to decide where to meet and cannot communicate with each other before making their choices. There are two possible locations: the Campus Café and the Library.
The game can be represented by the following payoff matrix:

		Bob	
		Café	**Library**
Alice	**Café**	6, 2	0, 0
	Library	0, 0	3, 4

The Library, however, has traditionally been the place where students in their program meet when no location has been specified. Alice and Bob are both aware of this tradition.

a. Find all Nash equilibria of this game.
b. Keeping in mind Schelling's idea of a focal point, which outcome do you think is the best prediction of where Alice and Bob will meet? Explain how the context of the problem affects your prediction.
c. Are all the Nash equilibria you identified in part (a) socially optimal? Explain your answer.
d. Someone claims, "The conventional equilibrium that Alice and Bob tend to play must also be the best outcome for everyone." Do you agree with this statement? Explain using your answers from parts (b) and (c).